Homework 1
Circuit Simulation of Basic CMOS Circuits

handout: 2023/03/02

due: 2023/03/23

EDA tools: HSPICE

Use HSPICE circuit simulator to perform the following measurements based on CIC 0.18um virtual process.

1. INV_XL

Perform the following measurements

- propagation delay (rise delay and fall delay)
- rise time/fall transition time of output waveforms
- power (dynamic and leakage)
- input capacitance (at each input)

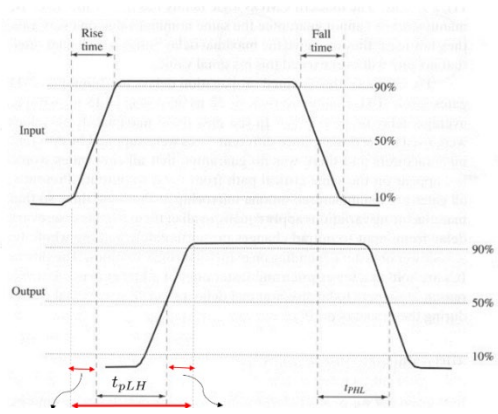
for the CMOS inverter circuit (named INV_XL) with minimum MOS transistor width $\mathbf{W_n=W_p=0.25um}$ at different load capacitances of $\mathbf{C_{load}=0,\ 0.01pF,\ 0.02pF,\ and\ 0.03pF}$ respectively. Use the minimum MOS length of L=0.18um for all the nMOS and pMOS.

Below shows an example of SPICE waveform of pulse. Use input slew of 0.03ns, i.e., tr = tf = 0.03ns, for all the simulation.

```
Vck clk gnd pulse 0 1.0 0ps 100ps 100ps 300ps 800ps
```

```
PULSE v1 v2 td tr tf pw per
```

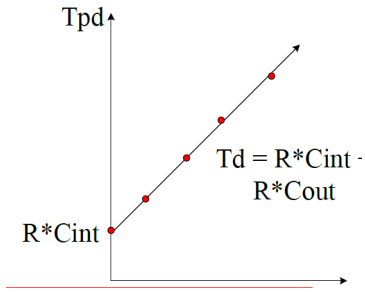
Definition of propagation delay is the time between Vin=0.5*Vdd and Vout=0.5*Vdd, as shown below.



The low-to-high propagation delay (T_{pLH}), also called rise delay, is the propagation delay of an output waveform from low to high due to some changes of inputs. Similarly, the high-to-low propagation delay (T_{pHL}), also called fall delay, is the propagation delay of an output waveform from high to low due to some changes of inputs. The rise time of an output waveform from low to high is defined as the time between 0.1*Vdd and 0.9*Vdd;

the fall time of an output waveform from high to low is defined as the time between 0.9*Vdd and 0.1*Vdd. Note the differences of rise/fall delay and rise/fall time where the rise time/fall time denotes the change rate of a "single" waveform while the rise/fall delay denotes the propagation delay delay of the output waveform with respect to the change of input waveform. The delay without external load is called *intrinsic delay* in the standard cell library data sheet.

In general, the total delay depends on input transition (slew rate) and output load capacitance. We will go into more details later when teaching the delay models. In the linear RC delay model to be addressed later in this course, the delay is proportional to the load capacitance, as shown below:



Let T_0, T_1, T_2, T_3 denote respectively the delays at different load capacitance C_i=0pF, 0.01pF, 0.02pF, 0.03pF, i=0, 1, 2, 3. We can calculate the 3 individual slopes $K_i = (T_i-T_0) / C_i$, i=1, 2, 3. The equivalent resistance can be approximated as the average $K=(K_1+K_2+K_3)/3$ in unit of ns/pF. Calculate the K values for the pull-up pMOS resitance and the pull-down nMOS resistance of the INV_XL. Compare the K values with those from the TSMC 180nm standard cell library data sheet (appended at the end of this homework assignment).

The power depends on the rate of "changes" of two neighboring input patterns. That is, power depends on frequency. Measure the power using randomly generated input period of 1us=1000ns so that the power is in unit of uW/MHz, equivalent to uJ, the unit of energy.

Complete the following tables along with the supporting results from the HSPICE simulation.

INV_XL (TT)	INV_XL @ Vdd=1.8V, Temp=25 C; input slew = 0.03ns input capacitance = ??? pF								
	propagation delay						transition time		power
load (pF)	rise (ns)	Ki (ns/pF)	K (ns/pF)	fall (ns)	Ki (ns/pF)	K (ns/pF)	rise (ns)	fall (ns)	(uW) @ 1MHz
0		-	-						
0.01									
0.02									
0.03									

2. Process Corners

Due to process variation, you might have different measurement results using different process corners (TT, SS, FF, FS, SF) as shown below and under different temperatures.

3.

For example, the default setup in problem 1 is processor corner TT (Typical nMOS/Typical pMOS) with room temperature 25 C . Redo problem 1 using another processor corner of SS. Compare the differences of these measurements with those in problem 1.

Complete the following tables along with the supporting results.

INV_XL (SS)	INV_XL @ processor corner of SS input capacitance = ??? pF								
	propagation delay						transition time		power
load (pF)	rise (ns)	Ki (ns/pF)	K (ns/pF)	fall (ns)	Ki (ns/pF)	K (ns/pF)	rise (ns)	fall (ns)	(uW) @ 1MHz
0		-	-						
0.01									
0.02									
0.03									

3. NAND2_XL

Redo problem 1 for CMOS two-input NAND gate circuit $Y = \overline{AB}$ with **W_n=W_p=0.25um and** L=0.18um for all MOS transistors. Name it as **NAND2_XL**. Note that the worse-case rise delay of CMOS two-input NAND is when the changes of input patterns and the corresponding output are 11/0 -> 10/1. The worse-case fall delay is when the logical inputs/output = 00/1 (or 01/1) -> 11/0. Why?

Complete the following tables along with the supporting results.

NAND2_XL (TT)	NAND2_XL @ Vdd=1.8V, Temp=25 C; input slew = 0.03ns input capacitance C_A= ??? pF, C_B= ??? pF								
	propagation delay						transition time		power
load (pF)	rise (ns) A->Y B->Y	Ki (ns/pF) A->Y B->Y	K (ns/pF) A->Y B->Y	fall (ns) A->Y B->Y	Ki (ns/pF) A->Y B->Y	K (ns/pF) A->Y B->Y	rise (ns)	fall (ns)	(uW) @ 1MHz
0		-	-						
0.01									
0.02									
0.03									

4. NOR2_XL:

Redo problems 1 for two-input CMOS NOR gate circuit $Y = \overline{A + B}$ with **W_n=W_p=0.25um and** L=0.18um **.** Name it as **NOR2_XL**. Hint: the worse-case rise delay of CMOS two-input NOR is when the logical inputs/output = 01/0 (or 11/0) -> 00/1. The worse-case fall delay is when the logical inputs/output = 00/1 -> 10/0. Why?

Complete the following tables along with the supporting results.

NOR2_XL (TT)	NOR2_XL @ Vdd=1.8V, Temp=25 C; input slew = 0.03ns input capacitance C_A= ??? pF, C_B= ??? pF								
	propagation delay						transition time		power
load (pF)	rise (ns) A->Y B->Y	Ki (ns/pF) A->Y B->Y	K (ns/pF) A->Y B->Y	fall (ns) A->Y B->Y	Ki (ns/pF) A->Y B->Y	K (ns/pF) A->Y B->Y	rise (ns)	fall (ns)	(uW) @ 1MHz
0		-	-						
0.01									
0.02									
0.03									

Format and Submission of homework report:

(1) Prepare your homework report as a complete MS word file containing all the necessary supporting information, figures and tables for the circuits and the simulation results, including schematic view, simulation waveform view, and HSPICE netlist file (*filename.sp*), etc.

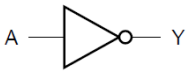
(2) Mark the propagation delays in your simulation waveform.

(3) Submit the homework by emailing the complete report file to TA with title: VLSI Homework 1 [student ID].

References:

Below lists some cell information from Artisan TSMC 0.18um (180nm) Standard Cell Library.

inverter (INV):



Functions

A	Y
0	1
1	0

Cell Size

Drive Strength	Height (μm)	Width (μm)
INVXL	5.04	1.32
INVX1	5.04	1.32
INVX2	5.04	1.98
INVX3	5.04	2.64
INVX4	5.04	2.64
INVX8	5.04	3.96
INVX12	5.04	8.58
INVX16	5.04	11.22
INVX20	5.04	12.54

AC Power

Pin	Power (μW/MHz)								
	XL	X1	X2	X3	X4	X8	X12	X16	X20
A	0.0087	0.0117	0.0218	0.0329	0.0394	0.0773	0.1706	0.2260	0.2820

Pin Capacitance

Pin	Capacitance (pF)								
	XL	X1	X2	X3	X4	X8	X12	X16	X20
A	0.0027	0.0036	0.0071	0.0104	0.0136	0.0271	0.0068	0.0090	0.0110

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)								
	XL	X1	X2	X3	X4	X8	X12	X16	X20
A → Y↑	0.0261	0.0253	0.0228	0.0243	0.0206	0.0198	0.1303	0.1276	0.126
A → Y↓	0.0154	0.0146	0.0140	0.0146	0.0125	0.0125	0.1235	0.1232	0.118

Description	K_{load} (ns/pF)								
	XL	X1	X2	X3	X4	X8	X12	X16	X20
A → Y↑	6.2539	4.5257	2.2629	1.5216	1.1447	0.5513	0.3680	0.2760	0.220
A → Y↓	3.3414	2.3675	1.2661	0.8247	0.6333	0.3211	0.2194	0.1647	0.131

two-input NAND (NAND2):

Logic Symbol

Functions

A	B	Y
0	x	1
x	0	1
1	1	0

Cell Size

Drive Strength	Height (μm)	Width (μm)
NAND2XL	5.04	1.98
NAND2X1	5.04	1.98
NAND2X2	5.04	3.30
NAND2X4	5.04	4.62

AC Power

Pin	Power (μW/MHz)			
	XL	X1	X2	X4
A	0.0106	0.0139	0.0258	0.0501
B	0.0135	0.0181	0.0351	0.0683

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A	0.0033	0.0042	0.0082	0.0163
B	0.0029	0.0039	0.0086	0.0158

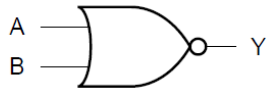
Delay at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)			
	XL	X1	X2	X4
A → Y↑	0.0324	0.0313	0.0286	0.0295
A → Y↓	0.0198	0.0195	0.0179	0.0181
B → Y↑	0.0409	0.0400	0.0389	0.0395
B → Y↓	0.0241	0.0242	0.0239	0.0235

Description	K_{load} (ns/pF)			
	XL	X1	X2	X4
A → Y↑	6.2614	4.5288	2.1944	1.1638
A → Y↓	3.8144	2.8429	1.3901	0.7200
B → Y↑	6.2527	4.5253	2.1935	1.1631
B → Y↓	3.8226	2.8470	1.3923	0.7210

two-input NOR (NOR2):

Logic Symbol



Functions

A	B	Y
0	0	1
x	1	0
1	x	0

Cell Size

Drive Strength	Height (μm)	Width (μm)
NOR2XL	5.04	1.98
NOR2X1	5.04	1.98
NOR2X2	5.04	3.30
NOR2X4	5.04	4.62

AC Power

Pin	Power (μW/MHz)			
	XL	X1	X2	X4
A	0.0110	0.0143	0.0275	0.0545
B	0.0139	0.0182	0.0365	0.0728

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
A	0.0033	0.0045	0.0083	0.0169
B	0.0029	0.0040	0.0085	0.0160

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)			
	XL	X1	X2	X4
A → Y↑	0.0426	0.0403	0.0347	0.0351
A → Y↓	0.0213	0.0196	0.0183	0.0187
B → Y↑	0.0536	0.0519	0.0485	0.0473
B → Y↓	0.0259	0.0244	0.0247	0.0245

Description	K_{load} (ns/pF)			
	XL	X1	X2	X4
A → Y↑	9.4704	6.7329	3.3656	1.6550
A → Y↓	3.5015	2.3672	1.2659	0.6330
B → Y↑	9.4589	6.7278	3.3647	1.6542
B → Y↓	3.5234	2.3779	1.2717	0.6357